Linearization

Introduction to CGE Modeling

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Outline

- The principle of linearization
- The rule of linearization
- Linearization with some key equations in CGE model

The principle of linearization

$$
\begin{aligned}
Y &= f\left(X_1, \ldots, X_n\right) \\
dY &= \sum_i \frac{\partial f\left(\cdot\right)}{\partial X_i} dX_i, \text{ total differentiation}
\end{aligned}
$$

- Define, the percentage change:

$$
\begin{aligned}
y &= 100\frac{dY}{Y}, \text{ and } x_i = 100\frac{dX_i}{X_i}, \text{ or} \\
dY &= \frac{Yy}{100}, \text{ and } dX_i = \frac{X_i x_i}{100}
\end{aligned}
$$

The principle of linarization (cont)

- By substitution:

$$\begin{aligned} Yy &= \sum_i \frac{\partial f(\cdot)}{\partial X_i} X_i x_i, \text{ intermediate change} \\ y &= \sum_i \frac{\partial f(\cdot)}{\partial X_i} S_i x_i, \text{ percentage change, where } S_i = \frac{X_i}{Y} \end{aligned}$$

The rule of linearization: power

$$Y =_i X_i^{\alpha_i}$$

- Applying $Yy = \sum_i \frac{\partial f(\cdot)}{\partial X_i} X_i x_i$:

$$\begin{aligned} Yy &= \sum_i \alpha_i X_i^{\alpha_i - 1}{}_{j \neq i} X_j^{\alpha_j} X_i x_i = \sum_i \alpha_i X_i^{-1}{}_i X_i^{\alpha_i} X_i x_i \\ &= \sum_i \alpha_i X_i^{-1} Y X_i x_i = \sum_i \alpha_i Y x_i \end{aligned}$$

- Divide by Y:

$$y = \sum_i \frac{\alpha_i Y x_i}{Y} \Rightarrow y = \alpha_i x_i$$

- Example:

$$Y = X^3 \Rightarrow Yy = 3x^3 x \Rightarrow y = 3x$$

The rule of linearization: sum

$$Y = \sum_i X_i$$

- Applying $Yy = \sum_i \frac{\partial f(\cdot)}{\partial X_i} X_i x_i$:

$$Yy = \sum_i X_i x_i$$

- Applying $y = \sum_i \frac{\partial f(\cdot)}{\partial X_i} S_i x_i$:

$$y = \sum_i S_i x_i, \text{ where } S_i = \frac{X_i}{Y}$$

The rule of linearization: sum

- Example:

$$
\begin{aligned}
Y &= X_1 + X_2 \Rightarrow Yy = X_1x_1 + X_2x_2 \\
y &= S_1x_1 + S_2x_2, \text{ where } S_1 = \frac{X_1}{Y} \text{ and } S_2 = \frac{X_2}{Y}
\end{aligned}
$$

The rule of linearization: fraction

$$Y = \frac{X}{M}$$

- Applying $Yy = \sum_i \frac{\partial f(\cdot)}{\partial X_i} X_i x_i$:

$$Yy = \frac{1}{M} Xx + \left(\frac{X}{-M^2} \right) Mm \Rightarrow Yy = \frac{X}{M} x - \frac{X}{M} m$$

- Divide by Y:

$$y = x - m$$

- Learn and be familiar with 15 rules of linearization from the handout of linearization rule.

Group exercise

Derive the intermediate and percentage change of the following:

1. $Y = (XMZ)^{\beta}$
2. $Y = \frac{\sum_i X_i}{\sum_j Z_i}$
3. $Y = X - M - Z$
4. $Y = \frac{X-M}{Z}$
5. $Y = (X - M)^{\alpha}$
6. $Y = XM - PZ$

CES input demand in percentage change

- CES input demand:

$$X_i = Q\delta_i^{\sigma}\left(\frac{W_i}{C}\right)^{-\sigma}$$

where:

$$C = \left(\sum_i \delta_i^{\sigma} W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

- Let's $\hat{M}$ percentage change of m
- $X_i = Q\delta_i^{\sigma}\left(\frac{W_i}{C}\right)^{-\sigma}$:

$$\begin{aligned} \hat{X}_i &= \hat{Q} + \hat{\delta_i^{\sigma}} + \widehat{\left(\frac{W_i}{C}\right)^{-\sigma}} = \hat{Q} - \sigma\widehat{\left(\frac{W_i}{C}\right)} = \hat{Q} - \sigma\left(\hat{W}_i - \hat{C}\right) \\ x_i &= q - \sigma\left(w_i - c\right) \end{aligned}$$

CES input demand in percentage change (cont)

- $C = \left(\sum \delta_i^\sigma W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$:

$$\begin{aligned}\hat{C} &= \frac{1}{1-\sigma}\left(\sum_i \widehat{\delta_i^\sigma W_i^{1-\sigma}}\right) = \frac{1}{1-\sigma}\left(\sum_i \frac{\delta_i^\sigma W_i^{1-\sigma}}{\sum_j \delta_j^\sigma W_j^{1-\sigma}}\widehat{\delta_i^\sigma W_i^{1-\sigma}}\right)\\ &= \frac{1}{1-\sigma}\sum_i \frac{\delta_i^\sigma W_i^{1-\sigma}}{\sum_j \delta_j^\sigma W_j^{1-\sigma}}(1-\sigma)\,\hat{W}_i = \sum_i \frac{\delta_i^\sigma W_i^{1-\sigma}}{\sum_j \delta_j^\sigma W_j^{1-\sigma}}\hat{W}_i\end{aligned}$$

- $S_j = \frac{\delta_j^\sigma W_j^{1-\sigma}}{\sum \delta_i^\sigma W_i^{1-\sigma}} = \frac{W_i X_i}{\sum_i W_i X_i}$ is cost share (see derivation in the handout):

$$\hat{C} = \sum_i S_i \hat{W}_i \Rightarrow c = \sum S_i w_i$$

CES input demand in percentage change (summary)

- Level form:

$$X_i = Q\delta_i^{\sigma}\left(\frac{W_i}{C}\right)^{-\sigma}$$

where:

$$C = \left(\sum_i \delta_i^{\sigma} W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

- Percentage change form:

$$x_i = q - \sigma\left(w_i - c\right)$$

where

$$c = \sum S_i w_i$$

CES input demand with technical change

$$\min_{X_i} \sum_i W_i X_i \text{ subject to } Q = \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$$

- Define $\tilde{X}_i = \frac{X_i}{A_i}$ and $\tilde{W}_i = A_i W_i$ therefore $\tilde{W}_i \tilde{X}_i = A_i W \frac{X_i}{A_i} = W_i X_i$, the problem can be rewritten as:

$$\min_{\tilde{X}_i} \sum_i \tilde{W}_i \tilde{X}_i \text{ subject to } Q = \left\{ \sum_i \delta_i \left(\tilde{X}_i \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$$

CES Input demand with technical change (cont.)

The solution (in percentage change):

$$\tilde{x}_i = q - \sigma\left(\tilde{w}_i - c\right), \text{ where } c = \sum_i S_i \tilde{w}_i$$

$$x_i - a_i = q - \sigma\left(w_i + a_i - c\right), \text{ where } c = \sum_i S_i\left(w_i + a_i\right)$$

Intermediate form of indirect tax rate

- Relationship between purchaser price and producer's (basic) price:

$$P^C = (1+T)\,P^B, \text{ where } T = \frac{R}{P^B Q} = \frac{R}{V^B}$$

where $V^B = P^B Q$ is the value of sales at producer's price.

- Percentage change:

$$\widehat{P^C} = (\widehat{1+T}) + \widehat{P^B}$$

$$\begin{aligned}
(\widehat{1+T}) &= 100\frac{\Delta(1+T)}{1+T} = 100\frac{\Delta T}{1+T} = 100\frac{\Delta T}{1+\frac{R}{V^B}} \\
&= 100\frac{\Delta T}{\frac{V^B+R}{V^B}} = 100\frac{V^B}{V^B+R}\Delta T \Rightarrow p^C = p^B + 100\frac{V^B}{V^B+R}\Delta T
\end{aligned}$$

- Query: Why do we need tax rate in ordinary change? not percentage?

LES demand in percentage change

- LES demand system:

$$X_i = \gamma_i + \frac{\beta_i}{P_i}\left(Y - \sum_j P_j \gamma_j\right)$$

- We can rewrite:

$$X_i = \gamma_i + X_i^L$$

where

$$X_i^L = \frac{\beta_i}{P_i} Y^L \text{ and } Y^L = Y - \sum_j P_j \gamma_j \text{ is the supernumary income}$$

LES demand in percentage change

- Percentage change:

$$x_i^L = y^L - p_i$$
$$x_i = \lambda_i x_i^L$$

where $\lambda_i = -\frac{e_{ih}}{\phi_h}$, see module on consumer's theory.